Ontology-Guided Screening for Undetected Dysglycemia: Evaluating Clinical Knowledge Integration Under Temporal Shift in NHANES

Manuscript draft — ClinOntoScreen research prototype. All numbers in this manuscript are read from `artifacts/research_results.json` produced by the protocol-tagged run (`protocol-v1`, commit `dc7fc37`; full provenance in `research_results.json → protocol` and `docs/DECISIONS.md`).

> Research prototype only. Not a diagnostic tool, not clinically validated, and not for medical decisions.

Abstract

Background. Clinical guidelines and tabular machine learning are usually developed and evaluated separately; few studies test whether encoding guideline content as executable, provenance-carrying knowledge changes downstream screening decisions, and almost none do so under a temporal shift that can break hard-coded thresholds.

Methods. We built ClinOntoScreen: an OWL/RDF ontology that encodes the ADA 2026 diabetes-screening recommendations as executable rules with guideline-section references, coupled to logistic regression and XGBoost models. Leakage-safe predictors (HbA1c, glucose, diabetes questionnaire answers and medication variables excluded) were derived from two public NHANES cycles: 2017–March 2020 pre-pandemic (historical development, n = 6,103, 28.9% screen-positive) and August 2021–August 2023 (locked temporal test, n = 4,260, 23.1% screen-positive). The outcome is HbA1c ≥ 5.7% among adults reporting neither prior diabetes nor prior prediabetes. The primary (confirmatory) endpoint was sensitivity of a hybrid ranking (XGBoost + ontology-rule score) minus XGBoost alone at a fixed 20% screening capacity, with a paired participant bootstrap (1,000 replicates). Exploratory experiments probed low-data regimes (E1), masked inputs (E2), model-family effects (E3), disagreement review (H3), and the published CDC/ADA prediabetes risk test as a baseline.

Results. The primary effect was null: hybrid 0.409 vs ML-only 0.412 at 20% capacity; in the prespecified direction (hybrid − ML-only) the difference was **−0.003 (95% CI −0.010 to +0.008)**, entirely within the design's smallest detectable difference (0.014). At 20% capacity the prevalence ceiling is 0.867; the best policy (logistic + ontology features) reached sensitivity 0.423 = 0.488 of ceiling, PPV 0.488, lift 2.12. All trained models outperformed the published CDC/ADA point score (AUPRC 0.460–0.463 vs 0.380; sensitivity@20% 0.407–0.423 vs 0.372). Ontology features helped the linear model consistently in direction (sensitivity@20% +0.009, CI −0.004 to +0.019) and gave the smallest degradation under 30% input masking, but no interval excluded zero. Rule-only ranking (sensitivity@20% 0.343) and disagreement-based review (precision 0.360 vs size-matched confidence review 0.400) did not add value.

Conclusions. Encoding guideline provenance into an executable layer is feasible and auditable, but on NHANES with full historical data the rule-derived features are largely redundant with the raw inputs for ranking. The honest headline is a well-powered null on the primary endpoint, with a consistent but unresolved advantage for guideline features in a linear, low-data and missing-input settings. All code, protocol, and artifacts are released.

Keywords: clinical ontology, screening, NHANES, temporal validation, fixed-capacity evaluation, leakage control

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1. Introduction

Diabetes screening is a resource-allocation problem before it is a prediction problem: practices can screen only a bounded fraction of their population, so a model's value at a fixed referral capacity is often more decision-relevant than its AUROC. At the same time, guidelines such as the ADA Standards of Care encode expert consensus that is auditable in prose but inert in software. Tabular machine learning can rank risk effectively, yet it silently discards that consensus; guideline rule engines keep the consensus but usually are not evaluated under the same strict temporal generalization tests applied to learned models.

ClinOntoScreen asks a narrow, testable question: **does encoding a documented subset of screening guidance as an executable, provenance-carrying ontology change screening decisions or robustness, relative to raw features, when both are evaluated on a genuinely later survey cycle?**

Contributions:

1. An OWL/RDF ontology whose executable rules carry `sectionReference` (which guideline section justifies the rule), `ruleExpression`, and `operationalizationStatus` (which criteria — family history, gestational diabetes, PCOS — are *described but not computable* from NHANES), so the knowledge layer is auditable rather than implicit.
2. A leakage-safe, cycle-harmonized feature contract (including activity minutes harmonized across the two cycles' incompatible physical-activity instruments) with an automated contract test.
3. A capacity-constrained evaluation protocol: sensitivity/PPV/lift at 10/20/30% referral budgets, always reported against the prevalence ceiling `min(1, capacity/prevalence)`; one pre-specified confirmatory comparison; everything else explicitly exploratory with paired-bootstrap uncertainty and a pre-registered smallest-effect-of-interest computed on historical validation *before* test access.
4. An honest temporal evaluation on the latest cycle (August 2021–August 2023), including a published-knowledge baseline (CDC/ADA prediabetes risk test) and three pre-specified stress experiments (low data, missing inputs, model family).

2. Related work

NHANES-based dysglycemia prediction. NHANES supports a large prediction literature; closest to us in evaluation discipline is Kumar's leakage-controlled, survey-weighted benchmark of ten models with rolling-origin temporal cross-validation and the same 2021–2023 holdout (Zenodo, 2026), which also includes the ADA risk score as a baseline — but contains no knowledge-representation component. Casacchia et al. (2024) externally validated an EHR-derived prediabetes model on 2017–2020 NHANES, showing that variable-availability mismatch alone can block validation — a hazard we handle explicitly. CardioMeta (2026) performs leakage-reduced multi-task cardiometabolic prediction with NHANES temporal validation and MIMIC-IV transfer, foregrounding calibration and reporting discipline; it likewise has no guideline layer.

Ontology- and guideline-based decision support. Pham and Soo (2026) combine community detection with a diabetes ontology on Pima/130-US-hospitals/NHANES data and report high discrimination (NHANES AUROC ≈ 0.91) with semantic-reasoning interpretability — but with lab-inclusive features, a random-split design, and no temporal holdout or guideline-section provenance. Broader ontology-based CDS work

encodes knowledge but rarely reports fixed-capacity utility or uncertainty against a learned baseline under distribution shift.

Selecting evaluation design. Random splits overstate performance under survey-cycle shift; capacity-constrained (top-k) reporting is standard in screening/rare-event retrieval but rare in clinical risk-model papers, where AUROC/PR alone hides the operational trade-off. Our ceiling-adjusted capacity reporting makes the achievable operating point explicit.

Capability comparison with the three closest papers

Dimension	**Pham & Soo 2026 (CEAI)**	**Kumar 2026 (CMPB, Zenodo)**	**CardioMeta 2026 (arXiv)**	**ClinOntoScreen (this work)**
Data	Pima, 130-US-Hospitals, NHANES	NHANES 2005–2016 dev, 2017–2020 + 2021–2023 holdouts	NHANES dev + 2017–2020 temporal; MIMIC-IV transfer	NHANES 2017–Mar 2020 dev; **2021–2023 locked temporal test**
Target	Diabetes status (lab-inclusive)	HbA1c ≥ 6.5%, no diagnosed diabetes	Multi-task cardiometabolic status	HbA1c ≥ 5.7% in adults reporting no diabetes/prediabetes
Ontology / knowledge role	Domain ontology + clustering for semantics/interpretability	None	None	OWL/RDF rules with guideline-section provenance; features, ranking and policy
Leakage control	Not reported	Central (leakage-controlled)	Central (leakage-reduced heads)	Forbidden-predictor contract + pipeline-internal transforms + CI test
Baselines	Conventional ML/clustering	ADA risk score, survey-weighted GLM, boosting, ensembles	Strong GBDT/neural tabular baselines	LR/XGBoost raw, generic-feature control, rule-only, hybrid, **CDC/ADA published score**
Primary metric	Accuracy/AUROC	Discrimination + calibration + decision curves	Macro-AUROC/AUPRC + calibration	**Sensitivity @ fixed capacity with prevalence ceiling**, paired bootstrap CI
Missing-input / low-data probes	No	Ablations (unweighted, random split)	Domain-shift transfer	E2 masked inputs (rules report not-assessable), E1 learning curve

Dimension	Pham & Soo 2026 (CEAI)	Kumar 2026 (CMPB, Zenodo)	CardioMeta 2026 (arXiv)	**ClinOntoScreen (this work)**
Limitation vs. us	Random split; lab-inclusive; no temporal test	No knowledge layer	No guideline content; different target	NHANES only; survey weights not used; cross-sectional outcome

3. Data

Sources. CDC/NCHS public-use XPT files (DEMO, DIQ, GHB, BMX, BPXO, BPQ, SMQ, PAQ, MCQ, HDL) for NHANES 2017–March 2020 pre-pandemic and August 2021–August 2023, downloaded from `wwwn.cdc.gov/Nchs/Data/Nhanes/Public/...` by the notebook (no redistribution).

Cohort flow (from `artifacts/data_qc.json`).

Step	Historical (2017–Mar 2020)	Temporal test (2021–2023)
Rows in DEMO file	15,560 (merged panel)	11,933 (merged panel)
Adults 18+ → no reported diabetes → no reported prediabetes → valid HbA1c	6,103	4,260
Screen-positive (HbA1c ≥ 5.7%)	1,766 (28.94%)	983 (23.08%)
Secondary: HbA1c ≥ 6.5% in test cohort	—	67

Composition (Table: `artifacts/cohort_composition.csv`). Test cycle is older (mean age 49.2 vs 46.1; 36.7% vs 27.5% aged 60+) and less non-Hispanic Black (11.1% vs 24.7%) and non-Hispanic Asian (5.4% vs 11.8%) — a genuine covariate shift that, with the instrument changes below, defines the temporal-generalization challenge.

Cycle differences and harmonization. The physical-activity instruments changed completely (PAQ650/665 + PAD660/675 in the earlier cycle vs. leisure minutes PAD800/820 in the later one); they are harmonized into `leisure_activity_minutes` (skip-pattern zeros honored, 7777/9999/77777/99999 masked) and `sedentary_minutes` (PAD680, both cycles). `HDL` (LBDHDD) is available in both cycles and is included. Not harmonized (excluded from the schema and disclosed): `BPQ040A` hypertension-treatment item and `MCQ300C` family history exist only in the earlier cycle; cycle-specific weight columns (`WTMEC2YR`/`WTMECPRP`); a fourth BP reading. **Consequence: `hypertension_treatment` is 100% missingP in the test cohort** (verified in `data_qc.json`); other missingness is small (income-to-poverty ratio 12.5%, CVD history items ≈ 5%, waist 4.1%, BP readings 3.2%, HDL 4.5%).

Outcome. `LBXGH ≥ 5.7%` (prediabetes-range HbA1c) among adults answering "no" to both DIQ010 (told diabetes) and DIQ160 (told prediabetes). Single measurement, research label only — not a confirmed clinical diagnosis.

4. Ontology and rule engine

`ontology/diabetes_screening.ttl` (OWL/RDF, Turtle; 253+ triples; version 0.2.0) encodes an age-based screening rule and a younger-adult risk-based rule, criteria for age ≥ 35, overweight/obesity (25 kg/m² general; 23 kg/m² for non-Hispanic Asian participants, per ADA), hypertension, physical inactivity, CVD history, and higher-risk ethnicity, plus the screen-positive and diabetes-range outcome definitions. Every rule and outcome carries:

- `hasSource` / `ruleExpression` — machine-checked in CI;
- `sectionReference` — e.g. *ADA Standards of Care in Diabetes—2026, Chapter 2, screening recommendations for adults*;
- for criteria not operationalizable on NHANES (family history, gestational diabetes, PCOS): dedicated classes with `operationalizationStatus` marking them *described, not computed*.

The executor (`src/clinontoscreen/ontology.py` and the notebook equivalent) derives 11 `onto_*` features, returns *unknown* instead of guessing when inputs are missing (visible as `onto_criteria_missing_count`), and produces a per-participant rule-priority score. The ontology text embedded in the notebook is byte-identical to the repository file (contract test).

5. Methods

Leakage contract. Predictors exclude `LBXGH`, glucose, diabetes questionnaire answers, diabetes medication variables, `SEQN`, and any target-derived column (automated `FORBIDDEN_FEATURE_TOKENS` assertions in code and in `tests/`). 9 numeric + 11 categorical raw features; imputation, scaling and one-hot encoding fit inside each pipeline fold.

Models. Logistic regression (L2, `C` ∈ {0.1, 1, 10}) and XGBoost (n_estimators ∈ {150, 300}, depth ∈ {2, 3}, lr ∈ {0.03, 0.07}); hyperparameters selected by 3-fold stratified grid search on historical data with a 80/20 stratified historical validation split (n = 1,221, 353 positives) used for all selection, including the hybrid alpha grid {0, 0.025, 0.05, 0.1, 0.15, 0.2, 0.3}. Six variants: LR/XGB × raw/ontology features, an XGBoost no-higher-risk-ethnicity-flag ablation, and an XGBoost **generic-feature control** (independently coded columns with identical semantics, used to check that any gain comes from the values, not the ontology plumbing; values asserted equal).

Policies. Model probability; probability + α · rule score (hybrid); ontology rule score alone; CDC/ADA prediabetes risk test point total. Final models refit on all historical data, XGBoost ensembled across 5 seeds (logistic regression deterministic — seed SD reported separately).

Evaluation. Test outcomes read once, after git tag `protocol-v1` (commit `d30f8f8`; executed commit `dc7fc37` differs only in reporting-bug fixes, documented in `docs/DECISIONS.md`). Metrics: AUROC, AUPRC, Brier; capacity tables at 10/20/30% with `sensitivity_ceiling = min(1, capacity/prevalence)`, fraction-of-ceiling, PPV, lift; paired participant bootstrap (1,000) for sensitivity differences and AUPRC/AUROC differences; smallest detectable difference (MDE = 1.96 · SE from historical validation, computed before test access): **0.0143** (hybrid), **0.0146** (LR), **0.0179** (XGB).

Confirmatory vs exploratory. Only H2 (hybrid vs XGB-only sensitivity @20%) is confirmatory. E1 (training fractions 5–100%, 3 subsamples each), E2 (MCAR masking 10%/30% of waist, systolic/ diastolic BP and hypertension history; ontology features re-derived from masked inputs — see §6.7 for the executed mask set), E3 (model family), H3 (top-k disagreement vs size-matched confidence-only review), subgroups, and the ethnicity-flag ablation are exploratory and labelled as such.

6. Results

6.1 Historical validation (selection data)

Model	AUROC	AUPRC	Brier
Logistic raw	0.7763	0.5419	0.1676
Logistic + ontology	0.7802	0.5486	0.1663
XGBoost raw	0.7797	0.5433	0.1663
XGBoost + ontology	0.7806	0.5437	0.1657
XGBoost + ontology (no ethnicity flag)	0.7785	0.5391	0.1665

Hybrid alpha selected: **0.10** (0.10/0.15/0.20 tied exactly on validation sensitivity/PPV; tie broken toward the smallest alpha). Alpha 0.10 found 151/353 positives at 20% capacity vs 145 for α = 0.

6.2 Temporal test discrimination

Model (5-seed ensemble)	AUROC	AUPRC	Brier	seed SD (AUROC/AUPRC)
Logistic raw	0.7723	0.4598	0.1509	– (deterministic)
Logistic + ontology features	0.7718	**0.4612**	0.1503	–
XGBoost raw	0.7689	**0.4629**	0.1507	0.0005 / 0.0007
XGBoost + ontology features	0.7678	0.4600	0.1506	0.0004 / 0.0018
XGBoost + ontology (no ethnicity flag)	0.7686	0.4619	0.1505	0.0004 / 0.0011
XGBoost + generic control features	0.7678	0.4600	0.1506	0.0004 / 0.0018
CDC/ADA risk test (published)	*0.7180*	*0.3803*	*n/a (points)*	—

Figure: `artifacts/roc_pr_curves.png`, `artifacts/calibration.png`.

6.3 Fixed screening capacity (prevalence 0.2308; 20% ceiling = 0.867)

Policy	Sens @10%	Sens @20% (found/983)	Sens @30%	PPV @20%	Lift @20%	Frac. of ceiling @20%
XGBoost raw	0.230	0.412 (405)	0.582	0.475	2.06	0.475
XGBoost + ontology features	0.225	0.409 (402)	0.578	0.472	2.04	0.472
Logistic raw	0.223	0.414 (407)	0.593	0.478	2.07	0.478
Logistic + ontology features	0.226	**0.423 (416)**	**0.596**	**0.488**	**2.12**	**0.488**

Policy	Sens @10%	Sens @20% (found/983)	Sens @30%	PPV @20%	Lift @20%	Frac. of ceiling @20%
Ontology-only rule score	0.196	0.343 (337)	0.475	0.396	1.71	0.396
CDC/ADA risk test (published)	0.199	0.372 (366)	0.520	0.430	1.86	0.430

Figure: `artifacts/capacity_sensitivity.png` (ceiling drawn as dashed line).

Every trained model beats the published risk test at every capacity, but no policy comes close to the 0.867 ceiling: the best policy achieves about half of what perfect ranking could achieve at this budget, i.e. substantial headroom remains at this feature set.

6.4 Primary endpoint (H2, confirmatory) — null

	Sensitivity @20%	Positives found
XGBoost only (ML policy)	0.4120	405/983
Hybrid (XGB + 0.10 · rule score)	0.4090	402/983
Difference, prespecified direction (hybrid − ML-only)	**−0.0031, 95% CI [−0.0100, +0.0079]**	(paired bootstrap, 1,000; SE 0.0046)

(`research_results.json` stores the same interval as ML-only − hybrid = +0.0031 [−0.0079, +0.0100]; the sign convention is documented in `docs/DECISIONS.md`.)

The interval crosses zero and lies entirely within the pre-registered MDE (0.0143): the design cannot resolve an effect of this size, and none is claimed. The ethnicity-flag ablation was likewise null in the prespecified direction (ablated hybrid − ML-only = −0.0051, 95% CI [−0.0119, +0.0053]). Figure: `artifacts/primary_effect_ci.png`.

6.5 E3 — does the knowledge layer help one model family more? (exploratory)

Comparison	Δ sensitivity @20% (95% CI)	Δ AUPRC (95% CI)
Logistic + ontology vs raw	**+0.0092 [−0.0038, +0.0185]**	+0.0014 [−0.0047, +0.0070]
XGBoost + ontology vs raw	−0.0031 [−0.0181, +0.0114]	−0.0029 [−0.0092, +0.0038]
XGBoost + ontology (no ethnicity flag) vs raw	+0.0000 [−0.0110, +0.0144]	—

All intervals cross zero and all point effects are below the MDE. The **direction** matches the prespecified hypothesis (threshold flags help a linear model at least as much as a flexible tree), but the data do not establish it. The generic-feature control is numerically identical to the ontology-feature variant for XGBoost

(AUPRC 0.4600 both; sensitivity@20% 0.409 both), showing the rule columns carry no information beyond their raw inputs for the tree model. Figure: `artifacts/e3_model_family_deltas.png`.

6.6 E1 — low-data learning curve (exploratory)

Ontology-minus-raw deltas (AUPRC, sensitivity@20%) by training fraction:

Fraction (n)	Logistic ΔAUPRC	Logistic Δsens	XGBoost ΔAUPRC	XGBoost Δsens
5% (305)	−0.0044	−0.0119	+0.0008	+0.0020
10% (610)	−0.0037	−0.0010	+0.0012	+0.0044
25% (1,525)	−0.0007	−0.0020	−0.0014	+0.0003
50% (3,051)	**+0.0024**	**+0.0061**	−0.0021	−0.0027
100% (6,103)	+0.0014	+0.0092	−0.0044	−0.0061

No consistent low-data advantage: the linear model's gain turns positive only at ≥50% data, the tree model's turns negative. Reported as measured. Figure: `artifacts/e1_learning_curve.png`.

6.7 E2 — missing-input stress test (exploratory)

Mean over 3 masked repeats:

Model	AUPRC @0%	AUPRC @30%	Δ	Sens@20% @0%	Sens@20% @30%	Δ
Logistic + ontology	0.4612	0.4597	−0.0015	0.4232	0.4218	**−0.0014**
Logistic raw	0.4598	0.4578	−0.0020	0.4140	0.4140	−0.0000
XGBoost raw	0.4629	0.4597	−0.0032	0.4120	0.4076	−0.0044
XGBoost + ontology policy (hybrid)	**0.4633**	**0.4608**	−0.0025	0.4090	0.4093	+0.0003

The hybrid policy keeps the highest AUPRC at every masking rate; the linear models are the most stable at capacity. The rule layer never saw information the ML system lacked: ontology features were re-derived from the masked inputs and reported `not-assessable` rather than imputing rules. The not-assessable fraction stayed at 0.004 because masked items are risk factors, not eligibility inputs. **Executed mask set (disclosure):** `waist_cm`, `systolic_bp_mean`, `diastolic_bp_mean` and `hypertension_history`. The experiment register had also listed the four legacy yes/no physical-activity items; those items were dropped during cycle harmonization (100% missing in 2021–2023), and the runner masks only mask-list columns that exist, so the activity inputs were not masked in this run. The mask list was corrected post-run for future reruns only; no recorded number changes (`docs/DECISIONS.md`). Figure: `artifacts/e2_missing_input.png`.

6.8 H3 — guideline disagreement vs size-matched confidence review (exploratory)

Top-20% ML set and top-20% rule set overlap in 477 participants → disagreement set of 750 (270 positives, precision 0.360). The 750 participants nearest the ML cutoff (confidence-only review) contain 300 positives

(precision 0.400) against cohort prevalence 0.231. The disagreement set is **not** enriched over simple confidence review; reviewing where the guideline and the model disagree is not supported by these data.

6.9 Subgroups (exploratory, descriptive only)

Selected rows at 20% capacity (all groups ≥ 50 participants and ≥ 10 positives; full table in `research_results.json → subgroups`):

Subgroup	n	Prev.	Sens (XGB raw)	Sens (XGB + ontology)
All	4,260	0.231	0.412	0.409
Age 18–39	1,537	0.094	0.517	0.524
Age 60+	1,562	0.354	0.295	0.293
Female	2,347	0.220	0.431	0.451
Male	1,913	0.244	0.388	0.373
Non-Hispanic Black	473	0.383	0.271	0.276
Hispanic	731	0.242	0.469	0.452

Sensitivity is far lower in the high-prevalence 60+ group simply because the fixed capacity is oversubscribed there; differences between rows are not interaction tests and are not powered.

7. Discussion

What the results support.

1. **The primary endpoint is null and the null is informative.** With the protocol frozen before test access, α selected on historical validation, and an MDE computed in advance (0.0143), the hybrid rule-score policy neither helps nor harms top-20% triage: hybrid − ML-only = −0.003 [−0.010, +0.008].
2. **Guideline provenance can be made executable and auditable.** Every rule carries its guideline section reference; criteria that NHANES cannot operationalize are declared rather than faked; unknown inputs produce "not assessable" rather than silent defaults. This is a contribution of engineering method, not of predictive performance.
3. **Learned models beat the published risk test on this cohort** at every capacity and by a large margin in ranking quality (AUPRC +0.08–0.083), while the published score remains fully transparent. Two of its items (gestational diabetes, family history) are unavailable in NHANES, which limits this comparison — disclosed, not hidden.
4. **Knowledge content, not ontology plumbing.** The generic-feature control matches the ontology-feature variant exactly for XGBoost: converting known rules into extra columns neither adds information the tree cannot reconstruct nor invents signal.

What the results do not support. No claim of improvement on the primary endpoint; no claim that rule features help in low data (direction inconsistent across families); no claim that disagreement review adds value; no clinical validity of any kind.

Why the null is plausible. Rule features are deterministic transforms of raw inputs the models already see; with 6,103 historical participants the trees reconstruct them and the linear model's remaining edge (+0.009 at

capacity, CI crossing zero) is below resolution. The places where knowledge could in principle dominate — sparse labels, missing inputs — show small consistent gains for the guideline features (E2 stability; LR direction), which is exactly where a learned model has less to reconstruct from.

Practical role of the knowledge layer. On this evidence, its value is provenance, audit and behavior under missingness — not ranking. Deployment-minded use should treat the ontology as the reporting/explanation substrate with the model supplying the score.

8. Limitations and ethics

- Single HbA1c measurement; the outcome is a screen-positive label, not confirmed disease.
- NHANES is a public, cross-sectional, complex-survey sample; **survey weights are not used**, so results are not population-representative estimates, and cycle design differences confound part of the temporal shift (test cohort is older and demographically different).
- The ontology operationalizes a bounded subset of ADA guidance; family history, gestational diabetes and PCOS criteria are declared unoperationalized, and `hypertension_treatment` is unavailable in the test cycle (100% missing) — a documented measurement gap the model must live with.
- Race/ethnicity enters in two ways (rule flag; Asian-specific BMI threshold). The no-flag ablation is disclosed as partial: the Asian BMI threshold and the raw predictor remain.
- Exploratory experiments and subgroups are not multiplicity-adjusted; all intervals are shown.
- No prospective or clinical evaluation; no patient data used; the Streamlit explorer must not be used with real patient information.

9. Conclusion

Under a pre-specified, temporally held-out evaluation on NHANES 2021–2023, adding an executable guideline-ontology layer to screening models produced a **null primary effect on sensitivity at 20% capacity** (hybrid − ML-only = −0.003, 95% CI −0.010 to +0.008, below the pre-registered smallest detectable difference), while all trained models clearly outperformed the published CDC/ADA risk test and the knowledge layer showed its only consistent value where data are scarce or inputs go missing. The most defensible empirical finding is therefore a boundary: on rich tabular data, guideline features are redundant for ranking but not for auditability — and any future claim that knowledge integration improves screening must be made against this protocol, at this capacity, with this ceiling.

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Calibration	`artifacts/calibration.png`
Capacity sensitivity (all policies + ceiling)	`artifacts/capacity_sensitivity.png`
Primary effect CI	`artifacts/primary_effect_ci.png`
E1 learning curve	`artifacts/e1_learning_curve.png`
E2 missing-input stress	`artifacts/e2_missing_input.png`

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E3 model-family deltas	`artifacts/e3_model_family_deltas.png`
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